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Introduction to Generative AI

Generative Agents

Outline

- Module Learning Objectives
- Foundations of Agents
- Possible Worlds Semantics
- Theories of Agency
- Generative Agents

Module Learning Objectives

- Explain the foundations of agent theory, including autonomy, intentionality, and attributes like rationality and veracity
- Analyze the roles and applications of generative AI agents, such as chatbots, multi-agent systems, and creative agentsDemonstrate agents
- Discuss grades of intentionality, from first-order to higher-order systems, and their relevance to AI reasoning
- Use modal logic to model beliefs, desires, and intentions in agents, including possible worlds semantics and accessibility relations
- Address ethical considerations in designing agents, focusing on fairness, transparency, and goal alignment

Introduction

- An **Agent** is an entity that perceives environment and acts upon it to achieve goals
- Operates without direct intervention, adapts to changes, and learns from experience
- Agents involve in distributed systems, AI, robotics, and complex problem-solving
  - Offers modularity, scalability, and flexibility in system design
- Applied to Internet navigation, decision support, robots, and multi-agent systems
- Generative AI agents can generate text, images, and sounds; as multiple agents, self-reflective agents, chatbots, virtual assistants, and creative and simulating agents
- **Multiple Agents** as several AI agents interact with each other within a shared env
  - Collaborate, compete, or coexist, leading to complex emergent behaviors
  - Essential for distributed problem-solving and decentralized control

Foundations of Agents

- Agents attributes include **autonomy**, **social ability**, **reactivity**, **pro-activeness**, and possibly
  - **Mobility** the ability of an agent to move around an electronic network
  - **Veracity** assumption an agent will not knowingly communicate false information
  - **Benevolence** assumption that agents do not have conflicting goals, and that every agent will therefore always try to do what is asked of it
  - **Rationality** assumption that an agent will act in order to achieve its goals, and will not act in such a way as to prevent its goals being achieved
- Intentionality in complex systems, trade-off
  - More we know the system mechanism, less we rely on intentional descriptions
  - Complex systems (e.g., computers), have practical intentional explanations
    - Example: Clicking an icon → Belief: "User desires to open an application."

Agent Theory - Intentional Systems

- Entities whose behavior can be explained or predicted by
  - **Beliefs**, **desires**, and **rationality**
- Ascribing intentional notion of **beliefs**, **desires** and **goals** (intentions)
- Use intentional notions
  - Simplifies the explanation of complex systems
  - Makes behavior prediction more intuitive than purely mechanistic explanations
- Attributing beliefs and desires to machines is valid when it helps to
  - Understand their structure
  - Predict behavior
  - Diagnose and repair functionality

Grades of Intentionality

- **First-Order** Intentional Systems
  - Entities that possess beliefs and desires but do not reason about of others
  - Belief: $b_1(x)$, Desire: $d_1(y)$ where x, y are properties of the world
- **Second-Order** Intentional Systems
  - Entities that reason about their own and others' beliefs and desires
  - Belief about belief: $b_2(b_1(x))$, Desire about desire: $d_2(d_1(y))$
  - Creates hierarchical reasoning, where the agent evaluates others' intentions
- **Higher-Order** Intentionality
  - Extends beyond second-order to arbitrary levels of reasoning
  - $b_n(\dots b_2(b_1(x)) \dots)$, $d_n(\dots d_2(d_1(y)) \dots)$

Example Light Switch

- Mechanistic explanation
  - A light switch transmits current when toggled
- Intentional stance
  - Treat the switch as having
    - A desire to transmit current
    - A belief that the user wants light
- Mathematical abstraction
  - Action A: Transmit Current
  - Condition: $b(\text{User Wants Light}) \wedge d(\text{Light On})$

Core Components of Intentional Agents

- **Information Attitudes** Beliefs (B), Knowledge (K)
- **Pro-Attitudes** Desires (D), Intentions (I), Obligations (O)
- Minimal Agent Representation
  - An agent must have at least one information attitude and one pro-attitude
  - $A = (B, I)$ where $B \subseteq \text{World Knowledge}$, $I \subseteq \text{Desired Goals}$
- Intentionality offers a convenient abstraction for understanding agent behavior
- Rational agents link **beliefs** (information) to **desires** (actions)
  - Action A is taken if $B \Longrightarrow D$
- While intentionality is necessary for agenthood, it might not be sufficient

Logic and Intentional Notions

- Syntactic Problem in First-Order Logic
  - Treats formulas as truth-functional: the meaning of a compound expression depends only on the truth values of its components
  - Intentional notions like belief are referentially opaque, meaning substitution of equivalents doesn't preserve meaning
- Syntactic problem example, given, Bel(Olivia, Father(Zeus, Cronos))
  - If Zeus = Jupiter, substituting Jupiter gives Bel(Olivia, Father(Jupiter, Cronos))
    - This is intuitively invalid because Olivia might not equate Zeus and Jupiter
- Semantic Problem
  - Intentional notions are not truth-functional, e.g., Olivia believes p
  - The truth of believes p doesn't depend solely on p's truth value

Alternative Syntactic Approaches

- **Modal Logic** introduces modal operators (e.g., $\Box$ for necessity, $\Diamond$ for possibility)
  - Examples, $\Box(2+2=4)$, $\Diamond(\text{It rains tomorrow})$
  - Also read $\Box$ as "always" and $\Diamond$ as "sometime" (in the future)
- Intentional notions $\text{Bel}(a, \phi)$: "Agent a believes ϕ"
  - Allows direct manipulation of belief statements while avoiding referential opacity
- Agent A believes B if $\Box B_A$, A desires D if $\Diamond D_A$, A intends I if $\Box I_A$, and I aligns with D
- **Meta-Language Approach** uses a many-sorted first-order language with terms representing formulas in an object language
  - Beliefs are expressed in the meta-language
    - e.g., Bel(Olivia, **Formula**(Father(Zeus, Cronos)))

Alternative Semantic Approaches

- Possible Worlds Semantics models beliefs as sets of **accessible possible worlds**
  - Let W be the set of all possible worlds
  - $B_a(w) \subseteq W$ is the set of worlds a believes possible in world w
- Belief Condition $\mathrm{Bel}(a, \phi) \Leftrightarrow \forall w' \in B_a(w),\ w' \vDash \phi$
- Supports a robust correspondence theory
- **Logical Omniscience Problem** Agents are assumed to know all logical consequences of their beliefs
- **Sentential Models** as beliefs as explicit symbolic structures in the agent's memory
  - Example, an agent believes ϕ if ϕ exists in its belief database
- Simple and effective for practical reasoning systems
- Does not capture inferential capabilities directly

Mathematical Representation of Belief

- **Possible Worlds Formalism** (PWF) an agent a believes ϕ in a world w
  - $\mathrm{Bel}_a(\phi) \Leftrightarrow \forall w' \in R(w),\ w' \vDash \phi$
  - $R(w)$ is accessibility relation defining belief worlds from w
  - $w' \vDash \phi$: ϕ is true in world w'
- **Sentential Model** where belief is determined by the presence of symbolic data
  - $\mathrm{Bel}_a(\phi) \Leftrightarrow \phi \in \mathrm{Database}_a$
- PWF is theoretically rich but computationally complex for real systems
- Note: $\vDash$ is read as "entails," "logically implies," or "is true in all models of"
  - Semantic entailment, $\Gamma \vDash \phi$: "If all statements in Γ are true, then ϕ must also be true in every model"
  - Example, $P \wedge Q \vDash Q$: "From $P \wedge Q$, it follows that Q is true in all models"

Possible Worlds Semantics

- Possible Worlds Models (PWM)
  - Proposed by Hintikka (1962) and formalized by Kripke (1963)
- An agent's knowledge or beliefs are modeled as a set of possible worlds consistent with the agent's information
  - Epistemic alternatives are worlds the agent considers possible, given their beliefs
- Example
  - A poker player with the ace of spades eliminates possible card distributions inconsistent with this knowledge
  - The remaining configurations (worlds) represent their epistemic alternatives
- Applications in AI Reasoning, multi-agent systems, game theory

Syntax and Semantics of Modal Logic

- Operators
  - $\Box\phi$: "Necessarily ϕ" — ϕ is true in all accessible worlds
  - $\Diamond\phi$: "Possibly ϕ" — ϕ is true in at least one accessible world
  - Duality $\Box\phi \Leftrightarrow \neg\Diamond\neg\phi$, $\Diamond\phi \Leftrightarrow \neg\Box\neg\phi$
- Accessibility Relation (R)
  - Defines which worlds are **reachable** or **accessible** from the current world
- Truth Conditions
  - $M, w \vDash \Box\phi \Leftrightarrow \forall w' \, (wRw' \Longrightarrow M, w' \vDash \phi)$
  - $M, w \vDash \Diamond\phi \Leftrightarrow \exists w' \, (wRw' \wedge M, w' \vDash \phi)$
  - Model M evaluates the truth of modal formulas ϕ under a specific interpretation
    - The truth of ϕ in a world w is written as: $M, w \vDash \phi$

Key Axioms and Accessibility Relations

- The axioms of modal logic correspond to properties of the accessibility relation (R)

Axiom	Formula	Accessibility Relation
T (Truth Axiom)	$\Box\phi \Longrightarrow \phi$	Reflexive (wRw)
D (Deontic Logic)	$\Box\phi \Longrightarrow \Diamond\phi$	Serial ($\forall w \exists w', wRw'$)
4 (Positive Introspection Axiom)	$\Box\phi \Longrightarrow \Box\Box\phi$	Transitive ($wRw' \wedge w'Rw'' \Longrightarrow wRw''$)
5 (Negative Introspection Axiom)	$\Diamond\phi \Longrightarrow \Box\Diamond\phi$	Euclidean ($wRw_1 \wedge wRw_2 \Longrightarrow w_1Rw_2$)

Epistemic Logic for Knowledge and Belief

- Knowledge ($K_i\phi$)
  - An agent i knows ϕ if ϕ is true in all worlds accessible to i
  - $M, w \vDash K_i\phi \Leftrightarrow \forall w' \, (wRiw' \Longrightarrow M, w' \vDash \phi)$
- Belief ($B_i\phi$)
  - Similar to knowledge but does not require ϕ to be true
  - $M, w \vDash B_i\phi \Leftrightarrow \forall w' \, (wR_iw' \Longrightarrow M, w' \vDash \phi)$
- Multi-Agent Systems
  - R_i: Accessibility relation for agent i
  - $K_i\phi$: "Agent i knows ϕ"

Logical Omniscience Problem

- Necessitation Rule
  - If ϕ is valid ($\vDash \phi$), then $K_i\phi$ holds for all agents
  - This implies agents know all tautologies, which is unrealistic
    - A **tautology** is a logical statement that is true in all possible interpretations, such as $p \lor \neg p$ ("either p is true, or p is not true")
- Closure Under Logical Consequence
  - $K_i\phi \land (\phi \Longrightarrow \psi) \Longrightarrow K_i\psi$
  - This means agents know all consequences of their knowledge, which is counterintuitive for resource-bounded agents

Distinction btw Knowledge and Belief

- Knowledge
  - Satisfies T: $K_i\phi \Longrightarrow \phi$ (what is known is true)
  - Captures true beliefs
- Belief
  - Does not require T, so $B_i\phi$ may hold even if ϕ is false
  - Captures subjective reasoning

Alternatives to Possible Worlds Model

- Motivations for Alternatives
- The **logical omniscience problem** in possible worlds semantics (agents know all valid formulae and all logical consequences of their beliefs) motivates alternative approaches
- These alternatives attempt to model **resource-bounded reasoning** and address practical limitations

Levesque's Explicit and Implicit Beliefs

- Distinction between
  - **Explicit Beliefs** small, finite set of beliefs explicitly represented by the agent
  - **Implicit Beliefs** infinite set of beliefs logically entailed by explicit beliefs
- Two operators
  - $\mathrm{ExpBel}(\phi)$: Explicit belief in ϕ
  - $\mathrm{ImpBel}(\phi)$: Implicit belief in ϕ
- Explicit beliefs use a weakened possible worlds approach based on situations
- Implicit beliefs use standard possible worlds
- Limitations
  - Does not naturally support nested beliefs
  - The concept of "situations" is abstract and sometimes more problematic
  - Unrealistic predictions about reasoning capabilities under certain conditions

Logic of General Awareness

- Fagin and Halpern's model
- Builds on Levesque's ideas but simplifies semantics to avoid some limitations
- Focuses on **general awareness** rather than specific situations
- Criticized for simplicity but contributes to practical modeling of bounded reasoning

Konolige's Deduction Model

- Models beliefs of AI systems as a **deduction structure** rather than possible worlds
  - A deduction structure $d = (D, r)$
    - D: Base set of formulae (explicit beliefs)
    - r: Set of inference rules (possibly incomplete)
  - Deductive Closure $\mathrm{close}(d) = \{\phi \mid D \vdash_r \phi\}$ where$D \vdash_r \phi$ means ϕ is derivable from D using rules r
  - Belief Semantics $[i]\phi \Leftrightarrow \phi \in \mathrm{close}(d_i)$ where d_i is the deduction structure of agent i
- Note $\Gamma \vdash \phi$ is syntactic entailment, reads as "From the set of premises Γ, ϕ can be formally derived"
- **Advantages** directly models resource-limited reasoning, well-suited for practical AI
- **Limitations** assumes agents always apply inference rules consistently

Meta-Languages Syntactic Modalities

- Use a meta-language to describe beliefs, properties of agents in an object language
- Example, meta-language formula
  - Bel(Olivia,Father(Zeus, Cronos))
    - Represents Olivia's belief about a formula in the object language
- **Strengths** enables reasoning about properties of beliefs
- **Weaknesses** meta-language approaches are prone to inconsistency
  - e.g., Montague's paradox, arises in formal semantics when combining self-reference with a system rich enough to express truths about its own statements
    - Certain systems allowing self-referential statements lead to inconsistencies
    - Example, $\psi \Leftrightarrow \neg\Box T(\psi)$, "The statement ψ is T iff it is not necessarily T"
  - Solutions for consistency exist but increase complexity

Comparisons of Models

Model	Strengths	Weaknesses
Levesque's Model	Distinguishes explicit vs. implicit beliefs	Limited nested beliefs; abstract "situations"
Fagin & Halpern	Simpler semantics for awareness	Less expressive than possible worlds
Konolige's Model	Practical for AI systems; explicit deduction	Assumes perfect application of inference rules
Meta-Language	Rich representation of belief properties	Inconsistency risks; more complex to manage

Mathematical Insights

- Deductive Beliefs
  - $\mathrm{Bel}_i(\phi) \Leftrightarrow \phi \in \mathrm{close}(D_i, r)$
    - Only derivable beliefs are represented
- Explicit vs. Implicit
  - $\mathrm{ExpBel}(\phi)$: Directly stored beliefs
  - $\mathrm{ImpBel}(\phi)$: ϕ derivable from ExpBel
- Meta-Language Semantics
  - $\mathrm{Bel}(i, \phi)$: Represents agent i's belief about ϕ as an object language formula
- $\Gamma \vdash \phi$ implies that ϕ is provable in the system (syntactic reasoning)
- $\Gamma \vDash \phi$ implies that ϕ is true in every model satisfying Γ (semantic reasoning)

Theories of Agency

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Goals of Theories of Agency

- A complete theory of agency must
  - Represent dynamic aspects of agency, such as changes in an agent's cognitive state over time
  - Define relationships between key attributes of agency
    - Information (e.g., beliefs, knowledge)
    - Pro-attitudes (e.g., goals, desires, intentions)
  - Show how these attributes influence an agent's actions in response to the environment
- **AI and Robotics** model rational agents reasoning/acting in dynamic environments
- **Multi-Agent Systems** design protocols for cooperation and conflict resolution
- **Game Theory** analyze strategies based on intentions and beliefs

Moore: Knowledge and Action

- **Focus** preconditions of action—what an agent must know to perform an action
- Framework
  - Combines modal logic for knowledge (K) with dynamic logic for action
  - Allows for incomplete information and actions to discover how to achieve goals
- Strength is groundbreaking use of logic to model reasoning and action
- Example
  - If $K(\text{door is locked})$, an agent must act (find key) to achieve unlock door

Cohen and Levesque: Intention

- **Focus** formalizing intentions as prerequisites for rational action and speech acts
- Key Properties of Intentions (based on Bratman 1987, 1990)
  - Intentions create problems agents must solve
  - Intentions filter incompatible goals
  - Agents track their intentions' success and retry if needed
  - Agents believe their intentions are possible
  - Agents do not believe their intentions are impossible
  - Under some conditions, agents believe they will achieve their intentions
  - Agents do not necessarily intend all side effects of their intentions

Intention

- Persistent Goals
  - An agent has a persistent goal ϕ if

$$(\mathrm{Goal}(\phi) \wedge \neg K(\phi)) \Longrightarrow (\mathrm{Drop}(\phi) \text{ if } K(\phi) \vee K(\neg\phi))$$

- Intentions derived from persistent goals, where actions aim to achieve goals ϕ
  - Example: $\mathrm{Intend}(a)$ means

$$\mathrm{PersistentGoal}(\phi) \wedge \mathrm{DoAction}(a) \Longrightarrow K(\mathrm{AboutTo}(a))$$

- Applications multi-agent dialogue, conflict resolution, cooperative problem-solving

Belief-Desire-Intention Architecture

- **Focus** A comprehensive framework for **beliefs**, **desires**, and **intentions**
  - Beliefs (B): Information the agent considers true
  - Desires (D): States the agent wants to bring about
  - Intentions (I): Desires the agent commits to achieving
- Formalization based on **branching time logic**
  - Belief, desire, and intention modeled as accessible worlds branching over time
  - $I_a(\phi)$: Agent a intends ϕ
  - $I_a(\phi) \Longrightarrow D_a(\phi) \wedge \neg B_a(\neg\phi)$
- **Extensions** include **social plans** and coordination in multi-agent systems

Singh: Rich Agent Logics

- **Focus** a formalism capturing
  - Intentions
  - Beliefs and knowledge
  - Communication
  - "Know-how" for action
- Framework
  - Uses **branching-time semantics** to model agent behavior over time
  - Includes extensive axiomatization for reasoning about actions and communication

Werner: General Model of Agency

- **Focus** integrates insights from
  - Economics
  - Game theory
  - Automata theory
  - Philosophy
- **Goal** A unified model of agency with broad applicability
- **Status** Conceptually rich, but properties remain underexplored

Wooldridge: Multi-Agent Systems

- **Focus** formal methods for specifying and verifying multi-agent system protocols
- Approach
  - Uses temporal belief logics (linear and branching time)
  - Models multi-agent systems as state transition systems
- **Applications** verifying cooperative protocols, such as negotiation and coordination

Key Insights and Formalization

- Persistent Goals (Cohen and Levesque)
  - $\text{PersistentGoal}(\phi) = (\text{Goal}(\phi) \wedge \neg K(\phi)) \Longrightarrow (\text{Drop}(\phi) \text{ if } K(\phi) \vee K(\neg\phi))$
- Belief-Desire-Intention (BDI)
  - $I_a(\phi) \Longrightarrow D_a(\phi) \wedge \neg B_a(\neg\phi)$
- Knowledge and Action (Moore)
  - $K(\text{Preconditions}) \Longrightarrow \text{Ability to Act}$

Generative Agents

- **Self-reflective Agents** can monitor and evaluate their own performance
- **Chatbots** are AI agents designed to simulate human-like conversations
- **Virtual Assistants** can perform tasks or services based on user commands
- **Creative Agents** can generate original content such as art, music, or literature
- **Simulating Agents** mimic complex systems or environments, they can model scenarios for research, training, or predictive analysis
- Generative agents are computational systems to simulate believable human behavior
  - A memory stream for storing detailed observations
  - Reflection mechanisms for higher-order reasoning
  - Planning frameworks for consistent and coherent action over time

Memory and Retrieval in Gen Agents

- Memory retrieval scoring is a mechanism to determine the relevance of memories in guiding the agent's behavior, especially used for reacting, reflecting, or planning
- The memory stream maintains a comprehensive record of experiences
  - **Observations** as events perceived by the agent
  - **Reflections** as high-level abstractions of memories
- Memory retrieval scoring

$$S_{\text{retrieval}} = w_R \cdot R + w_I \cdot I + w_V \cdot V$$

  - R is recency score, such as exponential function, $e^{-\lambda t}$
  - I is importance, rated in 1 to 10 scale using natural language prompts
  - V is relevance, as Cosine similarity between memory and query embeddings

Reflection Mechanism

- Reflection synthesizes observations and prior reflections into higher-level insights
  - **Questions** generated by analyzing recent memory
  - **Insights** inferred based on retrieved memories
- Relevance calculation for memory selection
  - $\text{cosine_sim}(M_{\text{embedding}}, Query_{\text{embedding}}) = \frac{M \cdot Q}{\|M\| \|Q\|}$
- Memory tree structure for reflection enables recursive abstraction

Planning and Reacting

- Planning in generative agents ensures coherent actions over time
  - **Plans** are generated recursively from high-level goals to detailed steps
  - **Real-time Adjustments** based on environmental observations
- Example
  1. High-level plan: "Work on a research paper"
  2. Decomposition: "Write intro (9:00-10:00), summarize findings (10:00-11:00)"
- Plan consistency score $C_{\text{plan}} = \sum \frac{\text{action coherence}}{\text{total actions}}$

Emergent Social Behaviors

- Generative agents exhibit emergent social behaviors through interaction
  - **Information Diffusion** is spreading news or updates among agents
  - **Relationship Formation** which agents build connections over time
  - **Coordination** are joint activities and synchronized events
- Network density $D = \frac{2 \cdot |E|}{|V| \cdot (|V| - 1)}$
- Information spread measured via diffusion paths

Future Directions and Challenges

- Improving memory retrieval algorithms for efficiency and accuracy
- Expanding real-time interactions with parallelized architectures
- Addressing biases inherited from language models
- Challenges
  - **Computational Cost** manage memory streams for multiple agents
  - **Robustness** mitigate hallucinations and ensuring coherence
  - **Ethical Considerations** avoid over-reliance and ensuring transparency

Improving Memory Retrieval Algorithms

- Future systems will likely use a multi-tier memory architecture
  - Episodic memory for raw logs of events
  - Semantic memory for structured summaries and conceptual maps
  - Working memory for short-term context for active planning
- Advanced retrieval methods might include learned attention mechanisms that prioritize memories based on emotional salience, social relevance, or predictive utility
- Integrating knowledge graphs and event ontologies could improve grounding and reduce hallucinations during reflection and action generation

Expanding Real-Time Interactions

- Scalability remains a major bottleneck
- For generative agents to engage in synchronous conversations, act in rich environments, or participate in simulations with hundreds of entities, we need
  - Efficient model serving with low-latency generation
  - Incremental inference that avoids recomputing full plans
  - Agent summarization—compressing inactive agents without losing their continuity
- Solutions may draw from multi-agent reinforcement learning, systems optimization, and even neuromorphic computing to simulate large, interactive societies efficiently

Ethical Considerations

- Ethics is foundational in generative agent design
  - **Over-reliance** users may trust agents more than warranted, especially if they exhibit social signals of confidence
  - **Goal alignment** agents must act in accordance with human values and institutional constraints—hard to enforce in open-ended generative systems
  - **Privacy and memory** if agents remember sensitive data or reproduce real individuals' behaviors, it raises serious privacy and consent issues
  - **Transparency** it should be possible to audit and intervene in an agent's reasoning process, especially in safety-critical applications.
- Agent ethics must go beyond static rules
  - involves dynamic meta-reasoning, introspection, and adjustment mechanisms to align with human expectations and legal norms

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